

Rising hot nights, persistent cold days, and an aging population:

thermal extremes and cardiovascular hospital burden in Hong Kong, 2013–2023

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Email: shenruff@connect.hku.hk | Laidlaw Scholars research project | Ecological time-series (exposures & denominators complete; HA outcomes pending)

1. Why this matters for human health

Climate change is already reshaping thermal exposure in subtropical cities. Overnight heat and daytime extremes can impair physiological recovery, while cold snaps continue to stress cardiovascular systems via blood pressure, viscosity, and autonomic tone. Earlier local daily studies (e.g. Goggins and colleagues) found a **cold-dominant** cardiovascular hospitalization profile. Whether contemporary heat—especially official **hot nights** ($T_{min} \geq 28^{\circ}\text{C}$)—now coexists with persistent cold risk in an **aging** population is the open human-health question.

- Dense, humid urban living → limited overnight cooling
- Ages 65–74 nearly doubled (2013–2023) → more vulnerable person-time
- Hospital burden, not only mortality, matters for adaptation planning

2. Research aim

Aim. Quantify official thermal extremes and population aging (2013–2023) as the climate-human exposure base for a planned ecological assessment of HA admissions for AMI, ischemic stroke, and hemorrhagic stroke among residents aged 35+.

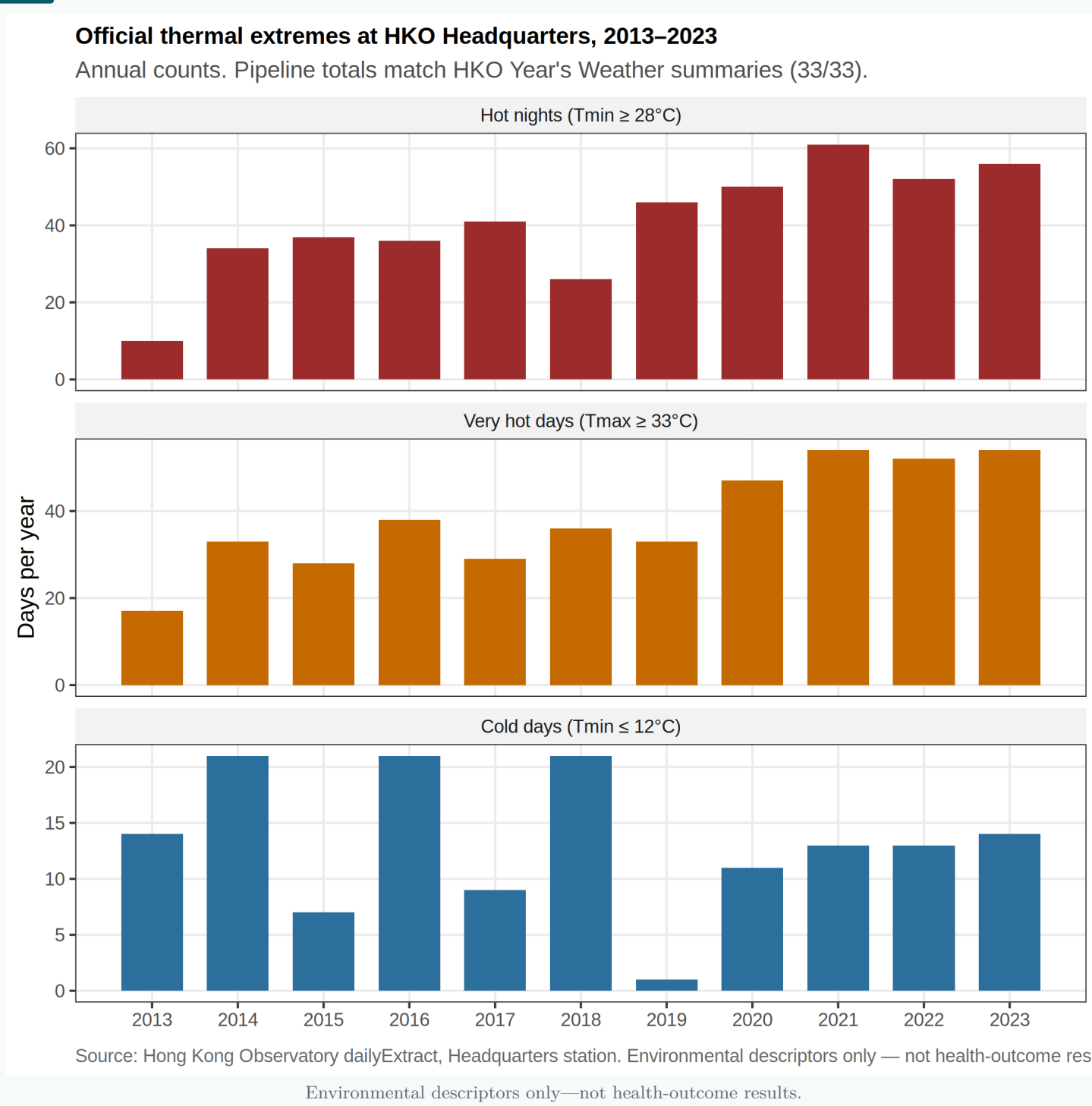
Design. Monthly ecological time-series ($N = 132$ months); population-level associations only.

Framing. Coexistence of emerging heat with persistent cold—not an assumed heat-for-cold shift.

3. Data & exposure definitions

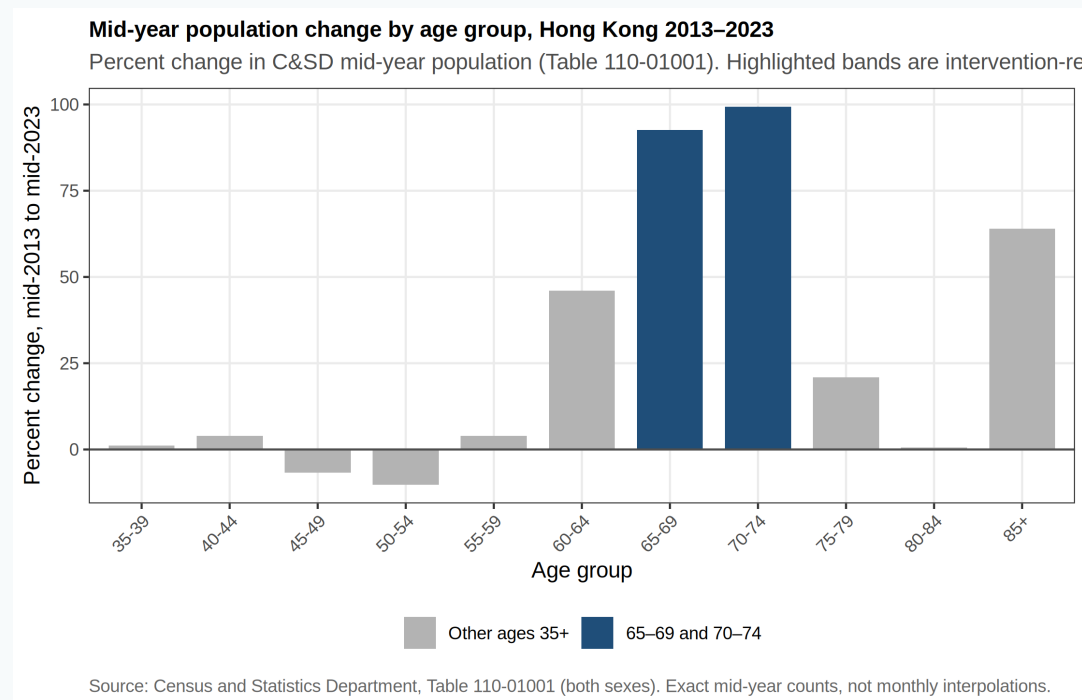
Domain	Source / status
Weather	HKO Headquarters daily extracts (REAL)
Population	C&SD Table 110-01001 mid-year age×sex (REAL)
Pollution	EFD series (import ready; staged)
Outcomes	HA CVD admissions (pending HPC access)
Official HKO extremes: hot night $T_{min} \geq 28^{\circ}\text{C}$; very hot day $T_{max} \geq 33^{\circ}\text{C}$; extremely hot day $T_{max} \geq 35^{\circ}\text{C}$; cold day $T_{min} \leq 12^{\circ}\text{C}$. Primary heat = monthly hot nights; co-primary cold = monthly cold days. Validation: annual totals 33/33 MATCH vs HKO <i>Year's Weather</i> .	

4. Rising heat, persistent cold (REAL)



- Hot nights: **10** (2013) → **61** (2021) / **56** (2023)
- Period means: **30.7** → **53.0** hot nights/year (2013–18 vs 2019–23)
- Cold days persist: **13–14** days/year in 2021–2023
- Hot nights in ≥ 5 -night spells: 34% → 51%

5. Aging raises the human stakes (REAL)



- Ages **65–69**: +92.6% | **70–74**: +99.3% | 85+: +63.9%
- More clinically vulnerable person-time under a warming subtropical climate

6. Seventeen analysis pathways

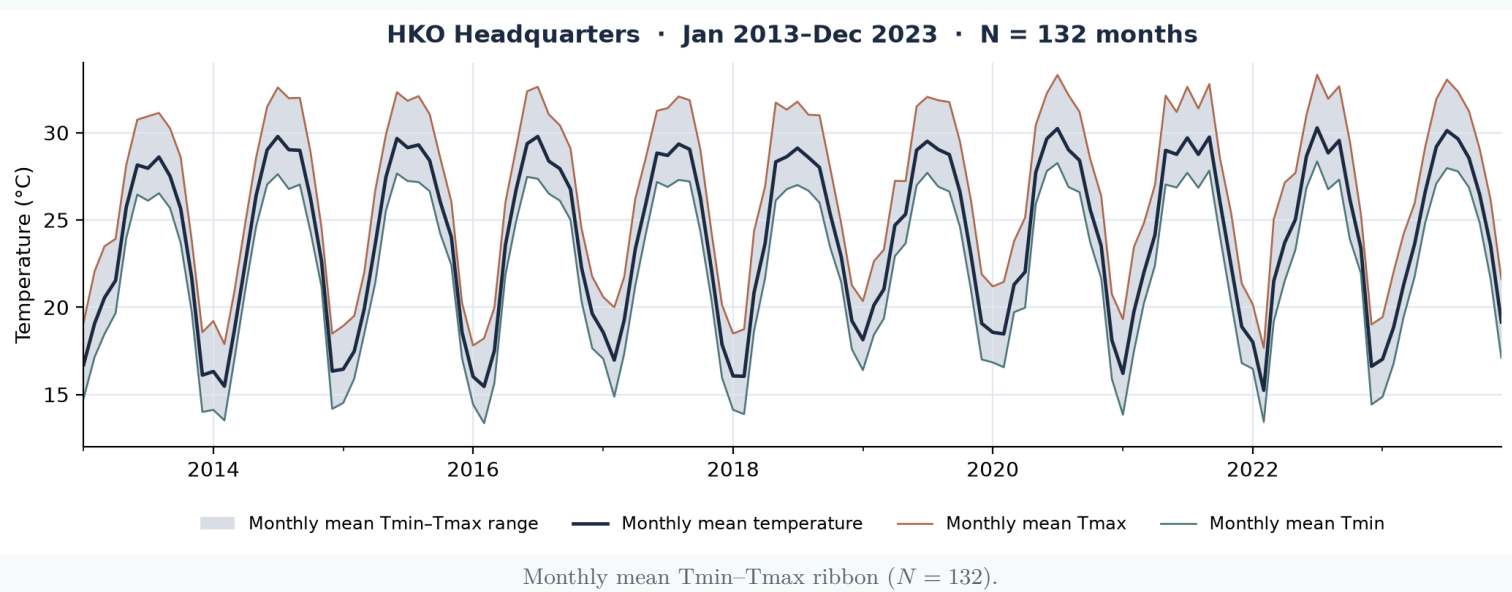
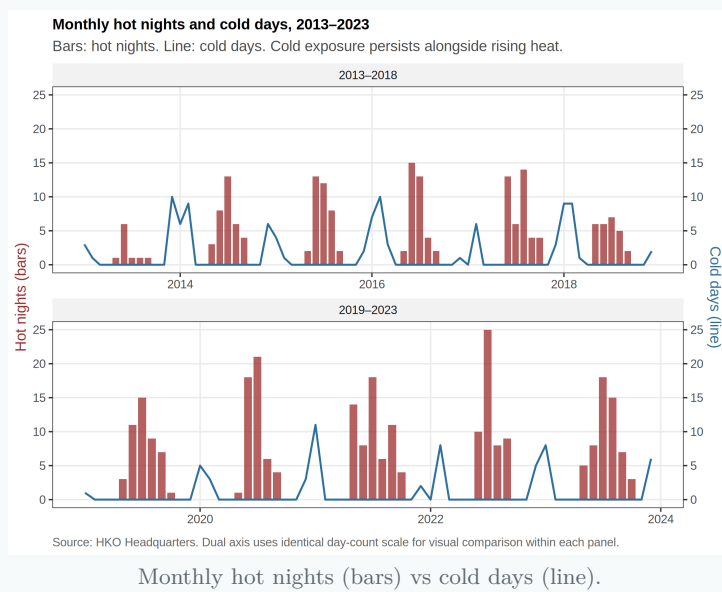
Numbered scripts turn public weather/population inputs into validated exposures, denominators, and (when HA arrives) admission models. Dev track: synthetic HA dry-runs only. Real track refuses syn-	
#	Pathway
00	Project setup & session info
01	Import HKO daily weather
02	Build monthly climate extremes
03	Import / document EFD pollution
04	Build monthly pollution panel
05	Build C&SD age×sex denominators
06	Confounders (COVID, holidays)
07	Synthetic HA dry-run outcomes
08	Merge analysis panel (dev)
08a	Merge approved HA panel
09	Descriptive tables
10	Exploratory figures
11	Fit NB models (syn, real)
12	Model diagnostics
13	Report outputs & status
14	Validated extremes figure
15	Smoke checks (HKO 33/33)
16	Exposure signing deliverable
17	Lab-meeting temp figures
Dev entry: run_pipeline_dev.R Real entry: run_pipeline_real.R Label: REAL / PLACEHOLDER / SYNTHETIC.	

7. Planned hospital-burden analysis

- Once HA access is live (ICD-9; patient-level admissions; ages 65–69 & 70–74 separable):
- Aggregate to month × age × sex × diagnosis
 - Negative-binomial rates; $\log(\text{population} \times \text{days})$ offset
 - Hot nights & cold days (per 5 extreme days); $T_{max}/T_{min} + \text{lag}$
 - Adjust for seasonality, trend, COVID phases; stage pollution
 - Outcomes: AMI, ischemic stroke, hemorrhagic stroke
- No synthetic model coefficients are presented as findings.

8. Take-home messages

- Climate is already changing Hong Kong's thermal environment: hot nights and very hot days rose sharply after 2018, while cold days did not disappear.
- Humans in the pathway: the near-doubling of ages 65–74 multiplies person-time at cardiovascular risk under heat and cold stress.
- A reproducible 17-step analysis platform is ready to merge official extremes with HA admission counts for transparent hospital-burden estimation.



9. Key numbers at a glance

10→61 Hot nights/year (2013 → 2021)	
30.7→53.0 Mean hot nights/year (2013-18 vs 2019-23)	
+93% / +99% Population ages 65-69 / 70-74	
33/33 HKO Year's Weather validation matches	
132 months Core window: Jan 2013-Dec 2023	
Next step: HPC onboarding → HA QC & Table 1 → pre-specified NB models. Contact: shenruff@connect.hku.hk School of Public Health, The University of Hong Kong GEST-2026 · Poster · Environmental Toxicology and Human Health	